AREPALLY VAMSHI KRISHNA

B.Tech.(Hons.) Metallurgical and Materials Engineering | IIT Kharagpur | CGPA: 7.49/10
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EDUCATION

Year	Degree / Exam	Institute	Score
2027	B.Tech.(Hons.) Metallurgical & Materials Engineering	IIT Kharagpur	7.49/10
2023	Higher Secondary (XII) — Science	Alphores Junior College	95%
2021	Secondary (X) — BSSE	Siddartha High School	10/10

PROJECTS

Personal Finance Dashboard | Python, Pandas, Matplotlib, NumPy, SQL **[Jan 2025]**

- Built 12-month personal finance tracker using Python and Pandas applying the 50/30/20 budgeting framework; used SQL for data querying and structured aggregation across categories
- Developed 5-panel Matplotlib dashboard: income vs expenses, savings rate trend, cumulative growth, expense pie chart, and 50/30/20 budget rule comparison with 20% target benchmark
- Automated CSV and PNG exports using NumPy statistical functions; identified 3% savings gap vs 20% target and modelled actionable budget reallocation strategies for improved financial health

Stock Portfolio Analysis | Python, NumPy, Pandas, Matplotlib, SQL **[Feb 2025]**

- Analysed mock Rs.1,00,000 NSE portfolio of 10 stocks using P/E ratio, EPS, Beta, Sharpe Ratio, Net Margin, Dividend Yield, and annualised volatility via NumPy return simulation
- Used SQL to store and query stock fundamentals; built 5-panel risk-return dashboard covering allocation, returns, P/E, sector P&L, and Beta scatter; identified IT sector as optimal allocation

3-Statement Financial Model — D2C Startup | Python, Pandas, SQL **[Mar 2025]**

- Built 5-year integrated P&L, Cash Flow, and Balance Sheet model for a D2C skincare startup using Python and Pandas with dynamic COGS, marketing, CapEx, and working capital assumptions
- Performed DCF valuation at 14% WACC with 4% terminal growth; break-even at 9,626 orders (Rs.81.8L); projected Year 3 as profitability inflection with gross margin expanding from 62% to 70%
- Modelled revenue scaling from Rs.1.02Cr (Year 1) to Rs.5.3Cr+ (Year 5); used SQL for structured financial data management; generated sensitivity analysis on key revenue and cost drivers

Fintech App Teardown + PRD | Product Management, RICE, Competitor Analysis **[Apr 2025]**

- Benchmarked CRED, PhonePe, Paytm, and Google Pay across MAU, NPS, UPI transaction volume, app ratings, load time, and 10 product features using a structured Python feature heatmap
- Applied RICE prioritisation framework (Reach, Impact, Confidence, Effort) to 8 feature candidates; Smart EMI Optimizer ranked #1 with highest reach-to-effort ratio on the product backlog
- Authored complete Product Requirements Document (PRD): problem statement, user persona, OKRs, user stories, and acceptance criteria; targets Rs.2,500/year savings and 5% D30 retention uplift

India BNPL & UPI Market Research | Python, SQL, Market Sizing, Opportunity Analysis **[May 2025]**

- Researched India's $38Bn BNPL ecosystem: 8 major players, 6 RBI/NPCI regulatory milestones, and UPI transaction growth from 10.8Bn (2019) to 172Bn (2024) with projected $94Bn GMV by 2027
- Built weighted opportunity scoring model using Python and SQL across 6 product areas: Market Size (35%), Feasibility (30%), Regulatory Risk (20%), and Competition (15%)
- Identified Embedded BNPL on UPI rails as the top product-market opportunity; delivered 4-panel Python visualisation dashboard covering market sizing, UPI growth, player landscape, and opportunity matrix

SKILLS AND EXPERTISE

Languages: C | C++ | MATLAB | Python | SQL | JavaScript | TypeScript | HTML | CSS
Tools & Platforms: Git | GitHub | VS Code | MongoDB | REST APIs | JWT Authentication | Netlify | Microsoft Excel
Libraries & Frameworks: NumPy | Pandas | Matplotlib | React | Next.js | Express.js
Finance & Product: Financial Modelling | DCF Valuation | Break-even Analysis | RICE Framework | PRD Writing | Market Sizing | Competitor Analysis | Portfolio Analysis | 50/30/20 Budgeting

RELEVANT COURSEWORK

Programming & Math: Programming and Data Structures | Probability and Statistics | Linear Algebra and Complex Analysis | Advanced Calculus | Transform Calculus | Computer Applications in Metallurgical Processes
Metallurgy: Thermodynamics and Kinetics | Deformation Behaviour of Materials | Materials Processing | Transport Phenomena **Others:** Electrical Technology | Engineering Mechanics | Physics of Waves

POSITIONS OF RESPONSIBILITY

Public Relations Team Member | Students' Alumni Cell, IIT Kharagpur **[Sep 2024 – Jun 2025]**

- Produced and edited video content for Instagram, YouTube, and LinkedIn; boosted student-alumni engagement and institutional visibility across digital platforms
- Collaborated with creative and event teams to deliver high-impact alumni event coverage; promoted student-alumni initiatives through structured cross-functional coordination

Secretary — Chess & Bridge | Nehru Hall of Residence, IIT Kharagpur **[Sep 2024 – Apr 2025]**

- Led Chess and Bridge activities: organised inter-hall competitions, mentored junior players, conducted introductory Bridge workshops, and represented hall in General Championship events

AWARDS AND ACHIEVEMENTS

- Ranked top 8% among 1.2M+ candidates in JEE Mains 2023; appeared for JEE Advanced 2023 — India's most competitive engineering entrance examination
- National Finalist — Product Construct 2025 (E-Cell IIT Madras); Finalist — INTSO Science Talent Search Olympiad (2018); AIR 242 — SilverZone Science Olympiad (2018)

EXTRA-CURRICULAR ACTIVITIES

- Mentored 5,000+ JEE aspirants in Physics and Mathematics at Physics Wallah, Expto, and RankersPrep; participated in L'Oreal Brandstorm 2025 Global Innovation Challenge
- Directed General Championship Short Film Making competition (IIT KGP); represented Nehru Hall in General Championship Volleyball 2024; volunteered with Vivekananda Welfare Association for the Blind